

Demo: Can AI Agents Adapt to Your Daily Life? HORAE, a Behavior-Aware Scheduler Grounded in Mobile Sensing

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Abstract

Modern scheduling tools manage fixed events but do not automatically adapt as a user's life changes—for example, when a user starts a scheduled activity later than planned, disrupting the rest of the day. Horae bridges this gap with a behavior-aware scheduler that adapts to evolving routines. Horae processes mobile sensor data directly on the user's device, learns what is normal for that person, and recommends simple schedule changes that better reflect changes in their life. In blinded evaluations, an AI reviewer preferred Horae's suggestions in 71.8% of cases. Human raters also preferred Horae, selecting its suggestions 57.3% of the time, compared with 36.5% for suggestions based only on existing calendar events and 6.3% for leaving the schedule unchanged. We demonstrate HORAE's end-to-end workflow live; a recorded video is available at: [Demo Link](#).

CCS Concepts

• **Computing methodologies** → **Intelligent agents**; *Natural language processing*; • **Information systems** → *Information retrieval*; • **Human-centered computing** → *Ubiquitous and mobile computing systems and tools*.

Keywords

Agentic AI, Large Language Models, Retrieval-Augmented Generation, Mobile Sensing

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1 Introduction

Although students lead increasingly fast-paced lives and struggle to balance their well-being with academic responsibilities, existing scheduling tools remain rigidly inflexible. Because traditional systems cannot adapt quickly enough to rapid behavioral changes—treating a 9 AM task the same

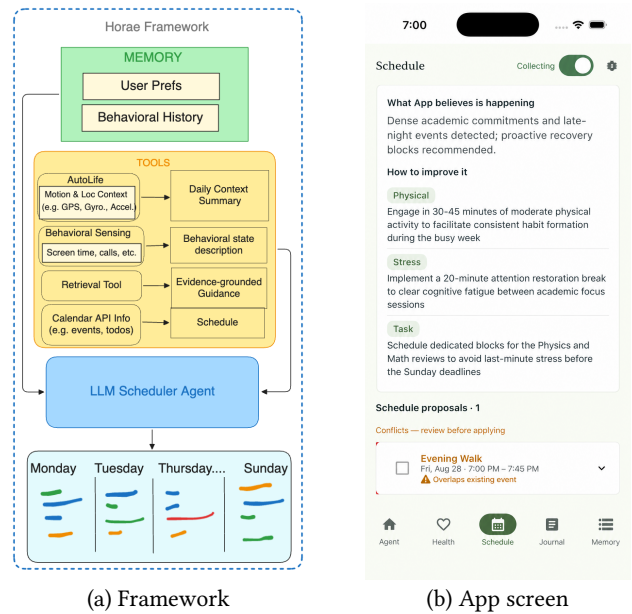


Figure 1: System and application views. (a) The HORAE framework. (b) The HORAE mobile app shows a behavioral summary, recommended actions, and a proposed calendar event with a conflict warning.



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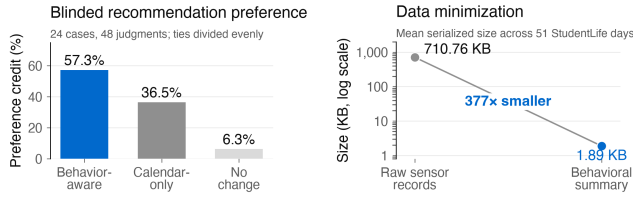


Figure 2: Preliminary results.

whether the user slept normally or went to bed late for several nights—the calendar strays further and further from the user’s actual daily life.

To bridge this gap, we present Horae, a dynamic, behavior-aware scheduler that adapts to a user’s evolving needs by combining passive mobile sensing [1, 5] with calendar context, user preferences, and behavioral-science evidence. Omitting redundant raw sensor data, the system utilizes existing framework models [6] and behavioral sensing to summarize daily habits and describe relevant routine shifts, such as changes in sleep and phone use.

Horae uses this compact behavioral context to generate a suitable schedule change—for example, suggesting an evening wind-down block in an available time slot. While earlier systems have summarized daily routines [2, 6] or used language models for calendar planning [3, 4], Horae connects these capabilities into a highly adaptable system where the user maintains control over adding each proposal.

2 System Design

Figure 1 shows the LLM scheduler connected to memory and four tools. On the user’s phone, AutoLife [6] converts raw motion and location records into a compact daily context summary. On the backend, a separate behavioral-sensing component derives behavioral markers from sleep and phone-use data, compares the most recent four days with up to 28 earlier days from the same user, and reports a possible routine shift together with the observations and data coverage supporting it. Raw records remain on the device; only the resulting summaries are passed to the rest of the system.

The LLM scheduler receives these summaries, upcoming events and pending tasks, user preferences and prior feedback from memory, and guidance from the VectorDB (a corpus of research papers about behavioral patterns and scheduling support). Using these inputs, the HORAE agent proposes a suitable calendar change that adapts to the user’s routine and fits their existing schedule. The user reviews the proposal before adding it.

3 Preliminary Results

We evaluated three questions: whether multi-day behavioral patterns relate to self-reported stress, whether human raters

prefer behavior-aware scheduling, and how much sensing detail is removed before information reaches the scheduler.

Behavioral signals and stress. Using StudentLife, a public dataset of smartphone sensing and self-reported wellbeing from college students [5], we analyzed 1,864 stress responses collected across 1,132 participant-days from 46 participants. When HORAE detected behavioral withdrawal—a multi-day pattern of less varied mobility and more time spent at frequently visited locations—90.9% of responses indicated stress, compared with 74.2% when the pattern was absent. After accounting for multiple responses from the same participant, stressed responses were still more common when behavioral withdrawal was detected.

Blinded recommendation comparison. We compared three scheduling approaches: (1) behavior-aware, which used both detected behavioral changes and the user’s calendar; (2) calendar-only, which used the same calendar but received no behavioral information; and (3) no-change, which proposed leaving the calendar unchanged. In blinded evaluations, an AI reviewer preferred Horae’s suggestions in 71.8% of cases. Two blinded human raters evaluated all three options across 24 cases (48 judgments) without seeing which approach produced each option. After dividing ties evenly, behavior-aware recommendations received 57.3% of the preference credit, compared with 36.5% for calendar-only and 6.3% for no change. Behavior-aware recommendations led when a sustained pattern or meaningful deviation was present, whereas calendar-only led when behavior was normal. While the small comparison provides preliminary preference evidence only, the result suggests that behavioral context may be most useful when a routine has meaningfully changed. The raters agreed on 62.5% of judgments.

Efficiency and On-Device Feasibility. In an offline analysis of 51 StudentLife days, the behavioral summary provided to the scheduler averaged 1.89 KB, compared with 710.76 KB for the corresponding raw sensor records—a 377× reduction. The scheduler receives derived behavioral indicators rather than the underlying raw records to limit data exposure and reduce downstream processing requirements. Rule-based inference takes ~1.7 ms per participant-day on a laptop CPU (excluding I/O, raw aggregation, and device runtime). Across three outcome-blind, duration-matched comparisons on a Pixel 9 and an iPhone 11, the median typical-minus-idle drain was +0.3 percentage points per hour on Android and 0.0 on iOS. These coarse, noncontemporaneous readings suggest small median battery overhead on the tested phones.

4 Demo Setup

Live setup. Visitors use a shared screen showing a weekly schedule and how HORAE forms a suggestion, step by step.

Fictional students mean that visitors do not share phone data or wait for the system to learn their usual routine. Each profile has prepared details about recent sleep and activity, calendar events, unfinished tasks, and personal preferences. Only these starting details are prepared; the summary and schedule suggestion are created live. The demo does not collect or retain visitor data.

What visitors see. The visitor chooses a profile and reviews the student's recent activity and calendar. After the visitor asks for a suggestion, the screen shows each step as it is completed: (1) an AutoLife summary of the student's day; (2) any change in sleep, movement, or phone use, together with how much information was available; (3) upcoming events and unfinished tasks; (4) relevant advice drawn from behavioral research; and (5) a suggested calendar change and whether it overlaps an existing event. The weekly schedule then shows the suggested event beside the student's existing plans. The visitor can read why it was suggested and decide whether to add it.

Changing a detail. The visitor can then change one detail, such as the student's sleep time, the amount of phone information available, a time marked as busy, or an unfinished task. When the demonstration runs again, HORAE repeats the analysis and checks the updated calendar; it may create a new suggestion, report no schedule change for a normal routine, or stop when there is not enough data. The screen shows which steps and final result changed, so visitors can see that the results are produced live rather than stored in advance. A recorded walkthrough, available at [Demo Link](#), provides a fallback if the internet or AI service is unavailable.

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